

SIDA-Guard: A No-Training Quality-Aware Reliability Layer for Deepfake Detection

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Abstract—Deepfake detection has become an important research topic due to the increasing realism of AI-generated and face-manipulated videos. Existing deepfake detectors usually output a binary prediction, such as real or fake. However, real-world videos are often visually degraded by lighting flicker, motion blur, ghosting, background noise, compression, and camera instability. These natural visual degradations may reduce the reliability of detector predictions, especially when a detector is applied outside clean benchmark settings. This project proposes SIDA-Guard, a no-training quality-aware reliability layer for deepfake detection. Instead of training a new detector, SIDA-Guard analyzes the visual quality of input videos and estimates whether a detector prediction may be unreliable. The proposed framework generates controlled perturbations on whole-frame, face-region, and background-region areas, records perturbation metadata, extracts whole-frame and fixed face region-of-interest (ROI) quality metrics, and converts these metrics into an interpretable quality risk score. Experimental results show that whole-frame metrics capture global degradation such as flicker and blur, while fixed face ROI metrics distinguish face-region degradation from background-only degradation. The final SIDA-Guard risk score assigns the highest risk to mixed whole-frame degradation and the lowest risk to clean videos, demonstrating its ability to provide interpretable reliability warnings for deepfake detection pipelines.

Index Terms—Deepfake detection, reliability estimation, video quality, visual degradation, face ROI, perturbation analysis.

I. INTRODUCTION

Deepfake videos are synthetic or manipulated videos in which a person’s face, identity, or expression is altered using artificial intelligence techniques. As generative models and face manipulation pipelines become more powerful, manipulated media are increasingly difficult to distinguish from authentic media. This creates risks for misinformation, online trust, identity misuse, and multimedia forensics. Deepfake detection has therefore become an important task in computer vision and media security.

Most existing deepfake detection systems focus on classification accuracy. Given an image or video, a detector predicts whether the input is real or fake. Many detectors achieve strong performance on benchmark datasets such as FaceForensics++ [1], the DeepFake Detection Challenge (DFDC) dataset [2], and Celeb-DF [3]. However, a binary prediction alone is not always sufficient in practical scenarios. Real-world videos often include natural visual degradations such as lighting flicker, motion blur, ghosting, camera motion, background

noise, and compression artifacts. These degradations may not be caused by malicious manipulation, but they can still affect the visual cues used by deepfake detectors.

This project is motivated by a practical question: the problem is not only whether a video is fake, but also whether the detector’s prediction can be trusted. If a real video is low quality and is incorrectly classified as fake, the result may create a false alarm. In real-world applications, such false alarms can be harmful because they may lead to wrong accusations, unreliable content moderation, or reduced trust in authentic media.

In many deployment settings, video content is first uploaded, compressed, re-encoded, or recorded under uncontrolled conditions before it reaches a detector. A detector may still output a confident score even when the input violates assumptions seen during training. This creates a practical gap between benchmark accuracy and operational reliability. A high classification score is useful only when the input quality is suitable for that detector. For this reason, quality-aware reliability estimation can be viewed as a safety layer around a detector rather than as a competing classification model.

This project is also motivated by resource constraints. Training a modern deepfake detector normally requires large datasets, careful preprocessing, and GPU resources. In contrast, a reliability layer based on explicit video quality metrics can be implemented with lightweight computer vision operations. Such a layer is easier to reproduce, easier to explain, and can be used with different detectors in the future. SIDA-Guard therefore focuses on interpretable measurement rather than end-to-end detector training.

Instead of training a new deepfake detector, this project proposes *SIDA-Guard*, a no-training quality-aware reliability layer. SIDA-Guard analyzes whether the input video contains visual quality problems that may reduce prediction reliability. The framework does not replace existing detectors. Rather, it can be placed before or alongside a detector to estimate whether the detector output should be accepted, treated with caution, or rejected for further verification.

The main idea of SIDA-Guard is to analyze visual degradation at two levels: the whole frame and the face region. Whole-frame quality metrics capture global degradations, such as full-frame flicker or blur. However, whole-frame metrics alone may hide local facial degradation because the face occupies only

part of the frame. Since many deepfake detectors rely heavily on facial cues, this project introduces fixed face ROI metrics. The face region is detected from a clean reference video, and the same face region is applied to all perturbed versions of the same video. This allows the system to compare the same facial area under different degradation conditions and reduce instability caused by repeated face detection.

The design of SIDA-Guard follows four practical requirements. First, the method should not require retraining a deep neural network. Second, the output should be interpretable, so that the user can understand why a video is considered risky. Third, the method should distinguish face-region degradation from background-only degradation. Fourth, the method should produce reproducible output files, including metrics, metadata, preview images, and final risk scores. These requirements guide the architecture and experimental design described in later sections.

The main contributions of this project are summarized as follows:

- We propose SIDA-Guard, a no-training reliability layer for deepfake detection.
- We design controlled perturbations for whole-frame, face-only, and background-only degradation.
- We combine whole-frame quality metrics with fixed face ROI quality analysis.
- We convert visual quality metrics into an interpretable quality risk score and warning decision.

The rest of this report is organized as follows. Section II reviews related work on deepfake datasets, detection methods, and robustness issues. Section III introduces the proposed SIDA-Guard framework. Section IV presents the experimental analysis. Section V concludes the report and discusses future work.

II. RELATED WORK

A. Deepfake Detection Datasets

Deepfake detection research has been strongly driven by benchmark datasets. FaceForensics++ is one of the early large-scale datasets for facial manipulation detection. It contains original videos and manipulated videos generated using multiple face manipulation methods, including DeepFakes, Face2Face, FaceSwap, and NeuralTextures [1]. It has been widely used to train and evaluate detectors under different compression settings.

The DeepFake Detection Challenge (DFDC) dataset was introduced to support large-scale research on face-swap video detection [2]. Compared with earlier datasets, DFDC includes more diverse subjects and recording conditions, making it useful for studying detector performance in more realistic environments. However, using DFDC for full-scale model training can require significant storage and computational resources.

Celeb-DF was proposed as a more challenging benchmark containing high-quality deepfake videos [3]. The dataset was

designed to address limitations of earlier datasets where generated videos could contain obvious visual artifacts. Celeb-DF therefore increases the difficulty of deepfake detection and provides a useful benchmark for evaluating detector generalization.

These datasets are important for training and benchmarking deepfake detectors. However, the goal of SIDA-Guard is different. Instead of training a detector on a large-scale dataset, this project focuses on whether visual degradation may reduce the reliability of detector predictions. This makes the framework lightweight and suitable for a short-term prototype without extensive training resources.

B. Deepfake Detection Methods

Deepfake detection methods commonly rely on convolutional neural networks (CNNs) to identify manipulation artifacts in face images or video frames. Xception is a widely used deep CNN architecture based on depthwise separable convolutions [4], and it has often been adopted as a strong baseline for facial forgery detection. MesoNet is another compact neural network designed specifically for facial video forgery detection, focusing on mesoscopic image features [5].

Other methods attempt to detect more general forgery traces. Face X-Ray, for example, represents whether an input face can be decomposed into blending from two image sources and highlights blending boundaries [6]. More recently, DeepfakeBench has provided a standardized benchmark platform with unified data processing, multiple datasets, evaluation protocols, and implementations of several detection methods [7]. Such benchmarks are important because inconsistent preprocessing and evaluation can lead to unfair comparisons.

Although these detectors and benchmarks improve deepfake classification, most systems still output a prediction regardless of input quality. In real-world use, a detector may produce a real or fake label even when the input video is visually unstable or degraded. SIDA-Guard addresses this gap by estimating the reliability of the input before fully trusting detector output.

C. Robustness and Visual Degradation

Real-world videos often contain degradation unrelated to manipulation, such as blur, illumination variation, compression, camera movement, and temporal instability. These degradations can change low-level image statistics and may be confused with manipulation artifacts. For example, a detector that relies on subtle texture patterns may be affected by motion blur, while a detector sensitive to brightness or color changes may be affected by flicker.

Robustness under distribution shift remains an important challenge in deepfake detection. Benchmark datasets are useful, but they cannot fully represent all real-world recording conditions. Therefore, reliability estimation is complementary to classification. Instead of forcing the detector to always make a decision, a reliability layer can warn users when the input is risky.

SIDA-Guard explicitly separates whole-frame degradation, face-region degradation, and background-only degradation.

TABLE I
POSITIONING OF SIDA-GUARD COMPARED WITH DETECTOR-CENTRIC WORK.

Work Type	Main Goal	Relation to SIDA-Guard
Datasets	Provide real/fake videos for training and testing	Used as possible input sources
Detector models	Classify videos or frames as real/fake	Can be combined with SIDA-Guard later
Benchmarks	Standardize detector evaluation	Useful future evaluation environment
SIDA-Guard	Estimate quality risk and prediction reliability	Adds warning before trusting detector output

This allows the framework to analyze where the quality problem occurs and whether it directly affects the facial region, which is often the main source of information for face-based deepfake detectors.

D. Quality Estimation and Reliability-Oriented Analysis

Image and video quality assessment has long been used to describe degradation such as blur, brightness change, noise, and temporal instability. In this project, these classical quality concepts are adapted to a deepfake reliability setting. The goal is not to judge aesthetic quality. Instead, the goal is to estimate whether a video contains visual conditions that may interfere with deepfake detection.

Reliability-oriented analysis is also related to abstention and warning mechanisms in machine learning. A model that always outputs a label may be inappropriate when the input is out of distribution or visually corrupted. A reliability layer can provide an additional decision state between acceptance and rejection. In SIDA-Guard, this idea is represented by three possible outcomes: Accept, Warning, and Abstain. This design is useful because it allows a detection pipeline to avoid over-trusting predictions under poor input conditions.

E. Positioning Compared with Existing Work

Most published deepfake detection work focuses on improving detector accuracy, dataset coverage, or benchmark fairness. SIDA-Guard focuses on a complementary question. It does not attempt to outperform Xception, MesoNet, Face X-Ray, or DeepfakeBench detectors. Instead, it asks whether the input video is visually reliable enough for a detector decision to be trusted. Table I summarizes this positioning.

III. PROPOSED APPROACH

A. Overview

SIDA-Guard is designed as a no-training reliability layer for deepfake detection. The system takes videos as input and analyzes whether visual degradation may make detector predictions unreliable. The full pipeline consists of five stages: video sampling, controlled perturbation generation, perturbation metadata recording, whole-frame and fixed face ROI metric extraction, and quality risk scoring.

Fig. 1 shows the conceptual architecture of the framework. Unlike conventional detectors, SIDA-Guard does not directly



Fig. 1. Overview of the proposed SIDA-Guard framework. The pipeline first generates controlled perturbations, extracts whole-frame and fixed face ROI metrics, and then converts the metrics into a reliability decision.

classify videos as real or fake. Instead, it produces a quality risk score and a reliability decision. This decision can be used together with any existing deepfake detector.

B. Design Objectives

The proposed framework is built for interpretability and reproducibility. Instead of learning hidden features from large-scale data, SIDA-Guard relies on explicit quality measurements. Each output score is linked to a measurable property, such as brightness fluctuation, sharpness loss, or frame-to-frame instability. This makes the result easier to explain in a final report or presentation.

Another objective is controlled comparison. If a video is degraded in many uncontrolled ways, it is difficult to determine which factor caused a reliability problem. Therefore, the pipeline creates controlled perturbation conditions. Each condition modifies a specific region and a specific type of degradation. This allows the analysis to compare clean, whole-frame, face-only, and background-only cases under the same processing structure.

C. Input and Output Organization

The implementation is designed to run in a Kaggle Notebook environment or a local Python environment. The input directory is searched recursively for supported video formats. Real and fake videos are identified using folder or filename keywords such as real, original, fake, deepfake, and synthesis. This keyword-based design allows the pipeline to work with different dataset folder structures without requiring a fixed dataset layout.

The output directory contains processed videos, metadata files, quality metric CSV files, preview images, summary tables, and final risk score files. This organization improves reproducibility because a user can inspect both intermediate and final outputs. Table II summarizes the main output files produced by the pipeline.

D. Controlled Perturbation Design

To analyze visual quality risk, SIDA-Guard generates controlled perturbations in three regions: whole-frame, face-only, and background-only. Whole-frame degradation modifies the entire video frame. Face-only degradation modifies only the facial region. Background-only degradation modifies the background while preserving the face region.

The perturbation conditions used in this project are summarized in Table III. Clean videos are used as the baseline.

TABLE II

MAJOR OUTPUT ARTIFACTS GENERATED BY THE SIDA-GUARD PIPELINE.

Output File or Folder	Purpose
processed videos	Clean and perturbed video variants
perturbation metadata	Perturbation type, region, frame range, and difference scores
quality metrics CSV	Whole-frame brightness, flicker, blur, and temporal metrics
fixed face ROI metrics	Face-region quality statistics across conditions
summary figures	Charts used for analysis and reporting
risk score CSV	Final quality risk and reliability score

TABLE III

PERTURBATION CONDITIONS USED IN SIDA-GUARD.

Condition	Region	Main Degradation
clean	none	none
light_flicker_whole	whole frame	brightness flicker
motion_blur_whole	whole frame	motion blur
ghosting_whole	whole frame	temporal ghosting
mixed_whole	whole frame	flicker + blur + ghosting
light_flicker_face	face	face brightness flicker
motion_blur_face	face	face blur
light_flicker_background	background	background flicker
motion_blur_background	background	background blur

Whole-frame degradation includes light flicker, motion blur, ghosting, and mixed degradation. Face-only degradation includes face flicker and face blur. Background-only degradation includes background flicker and background blur.

The mixed whole-frame condition combines flicker, motion blur, and ghosting. It simulates a more severe real-world video quality problem where several degradations occur at the same time. This condition is expected to produce higher risk than single perturbations because it affects multiple quality factors.

E. Perturbation Operators

The light flicker perturbation changes frame brightness over time. It is intended to simulate unstable lighting, such as indoor lighting variation, screen flicker, or changing exposure. A sinusoidal or frame-dependent brightness scaling factor is applied to the selected region. The motion blur perturbation applies a blur kernel to simulate camera movement or subject motion. The ghosting perturbation blends information from a previous frame into the current frame to simulate temporal residue or lagging visual trails.

For face-only perturbations, a face region is detected and used as the target mask. The degradation is applied only inside this face mask. For background-only perturbations, the face region is excluded and the degradation is applied to the remaining background. This mask-based design is important because it allows the experiment to test whether degradation outside the face affects face-level quality metrics.

The mixed whole-frame perturbation combines several operators in the same condition. It is intentionally more severe than a single perturbation. In real recordings, different quality problems often occur together; for example, a handheld video may

TABLE IV

EXAMPLES OF METADATA FIELDS RECORDED FOR EACH PROCESSED VIDEO.

Metadata Field	Description
video_id	Unique video identifier
label	Real or fake label from dataset structure
condition	Clean or perturbation condition
region	Whole frame, face, background, or none
start/end frame	Temporal interval where perturbation is applied
modified frames	Number of frames changed by the operator
ROI difference	Average difference in the intended target region
output path	Location of the generated video file

contain both motion blur and lighting instability. Therefore, mixed whole-frame degradation is useful for testing whether the final risk score increases under combined degradations.

F. Perturbation Metadata

For each processed video, SIDA-Guard records metadata including video identifier, label, condition, perturbation type, region, severity, parameters, start frame, end frame, affected frames, modified frames, face detection status, and ROI difference scores. Metadata is important because it makes the experiment traceable. Rather than simply generating degraded videos, the system records how and where each degradation was applied.

The ROI difference score compares the original frame and the perturbed frame in the target region. For whole-frame perturbations, the difference is computed over the entire frame. For face-only perturbations, the difference is computed only over the face region. For background-only perturbations, the face region is excluded and the difference is computed over the background area. This design helps verify whether the intended region was actually modified.

G. Whole-Frame Quality Metrics

Whole-frame metrics analyze the entire image sequence. The extracted metrics include brightness statistics, flicker score, blur score, saturation ratio, and temporal instability. The brightness score is computed as the mean grayscale intensity. The flicker score is computed as the standard deviation of brightness values over sampled frames:

$$F_{global} = \sigma(\mu_1, \mu_2, \dots, \mu_T), \quad (1)$$

where μ_t denotes the mean brightness of frame t and T denotes the number of sampled frames.

Blur score is computed using the variance of the Laplacian response:

$$B = \text{Var}(\nabla^2 I), \quad (2)$$

where I is the grayscale frame. A lower value indicates a blurrier frame because sharp edges are reduced. Temporal instability is measured using the average frame-to-frame difference between consecutive sampled frames:

$$T_{instability} = \frac{1}{T-1} \sum_{t=2}^T \|I_t - I_{t-1}\|_1. \quad (3)$$

These metrics allow SIDA-Guard to detect global degradation such as full-frame flicker and full-frame blur.

H. Fixed Face ROI Quality Metrics

Whole-frame metrics may hide local facial degradation because the face region may occupy only a small part of the frame. To address this issue, SIDA-Guard uses fixed face ROI analysis. The system first detects the face box from the clean video and then applies the same face box to all perturbed versions of the same video.

This design has two advantages. First, it allows the same facial area to be compared across conditions. Second, it avoids unstable face detection when videos are heavily degraded. If the face detector shifts or fails on a degraded video, dynamic face tracking would produce inconsistent face crops. Fixed face ROI reduces this problem.

The fixed face ROI metrics include face brightness, face flicker score, face blur score, face saturation ratio, and face temporal instability. These metrics estimate whether the facial region itself is degraded. This is particularly important for deepfake detection because many detectors focus on facial texture, blending boundaries, and facial inconsistencies.

I. SIDA-Guard Risk Score

SIDA-Guard computes two major risk components: global quality risk and face quality risk. Global quality risk is computed from whole-frame metrics, while face quality risk is computed from fixed face ROI metrics. For metrics where higher values indicate worse quality, such as flicker and temporal instability, risk is computed by comparing the current value with the clean baseline. For metrics where lower values indicate worse quality, such as blur score, risk is computed from the relative decrease compared with the clean baseline.

Let R_g denote global quality risk and R_f denote face quality risk. The final quality risk is computed as a weighted combination:

$$R_{final} = \alpha R_g + (1 - \alpha) R_f, \quad (4)$$

where α controls the contribution of global metrics. In this project, face risk is given higher importance because deepfake detectors often rely on facial regions.

The reliability score is defined as:

$$S_{reliability} = 1 - R_{final}. \quad (5)$$

In implementation, the risk score is normalized into the range from 0 to 1. A value close to 0 means that the video is similar to the clean baseline, while a value close to 1 means that the video contains severe quality degradation. The normalization uses the clean condition as the reference point, because the clean condition represents the expected low-risk case for each sampled video.

TABLE V
INTERPRETATION OF SIDA-GUARD RELIABILITY DECISIONS.

Risk Range	Decision	Meaning
Low	Accept	Input quality is likely reliable
Medium	Warning	Prediction should be treated carefully
High	Abstain	Further verification is recommended

Algorithm 1 SIDA-Guard Quality Risk Estimation

Require: Input video set V

Ensure: Quality risk score and reliability decision

- 1: **for** each video $v \in V$ **do**
- 2: Generate clean and perturbed conditions
- 3: Record perturbation metadata
- 4: Extract whole-frame metrics
- 5: Detect clean face box and build fixed face ROI
- 6: Extract fixed face ROI metrics
- 7: Compare metrics with clean baseline
- 8: Compute global risk R_g
- 9: Compute face risk R_f
- 10: Compute final risk R_{final}
- 11: Compute reliability score $1 - R_{final}$
- 12: Output Accept, Warning, or Abstain
- 13: **end for**

The decision thresholds are used as an interpretable reporting tool. Low risk corresponds to Accept, medium risk corresponds to Warning, and high risk corresponds to Abstain. The thresholds can be adjusted for different applications. For example, a strict forensic system may abstain more often, while a lightweight screening system may use a wider warning range.

Based on the final risk score, SIDA-Guard outputs one of three decisions: Accept, Warning, or Abstain. Accept indicates low quality risk, Warning indicates moderate risk, and Abstain indicates that the detector prediction should not be trusted directly without further verification.

IV. EXPERIMENTAL ANALYSIS

A. Experimental Setup

The experiment uses sampled real and fake videos. Each input video is processed into nine conditions: clean, light flicker whole, motion blur whole, ghosting whole, mixed whole, light flicker face, motion blur face, light flicker background, and motion blur background. With three real videos and three fake videos, the pipeline produces 54 processed videos.

For each processed video, SIDA-Guard generates a processed video file, preview image, perturbation metadata, whole-frame quality metrics, fixed face ROI metrics, and final risk scores. The experiment is designed as a prototype to validate whether quality risk can be measured and interpreted without training a new detector.

B. Evaluation Questions

The experimental analysis is organized around three evaluation questions. First, can the quality metrics detect the type

of degradation that was intentionally applied? Second, can fixed face ROI metrics distinguish face-region degradation from background-only degradation? Third, does the final risk score rank severe mixed degradation higher than clean or weak degradation conditions?

These questions are chosen because the project does not train a new detector. Therefore, the evaluation should focus on measurement validity and interpretability rather than classification accuracy. If the metrics correctly respond to controlled perturbations, then SIDA-Guard can provide meaningful reliability information for future detector integration.

C. Comparison Protocol

The analysis compares each perturbed condition with the clean baseline. This baseline comparison is important because different videos may have different natural brightness, sharpness, and motion characteristics. By comparing each condition against its corresponding clean reference, the system reduces the effect of video-to-video variation and focuses on the effect of the applied perturbation.

The experiment reports both numerical tables and visual charts. Tables provide exact values for representative conditions, while charts show overall trends across all perturbation types. Visual verification images are also included to show that the intended region was modified.

D. Visual Verification

Preview images are generated in the form of Original, Perturbed, and Difference. These visual comparisons verify whether the intended region was modified. In face-only perturbations, the difference map should mainly appear around the face. In background-only perturbations, the face region should remain mostly unchanged. Fig. 2 shows representative visual verification examples from the generated results.

This step is important because it confirms that the perturbation is controlled. It also helps explain the experiment visually, making it clear whether the degradation occurred in the whole frame, face region, or background.

The difference map is especially useful for detecting mistakes in the perturbation pipeline. If a face-only perturbation accidentally changes the background, the difference map would show changes outside the face. If a background-only perturbation accidentally changes the face, the difference map would show unwanted face-region differences. Therefore, visual verification acts as a qualitative sanity check before interpreting the numerical results.

E. Whole-Frame Metric Results

Whole-frame flicker score increases under lighting flicker conditions, especially when the entire frame or background is affected. This indicates that global brightness instability is successfully captured. Whole-frame sharpness score decreases under motion blur conditions. Since lower sharpness score indicates stronger blur, this confirms that motion blur perturbations are detected by the quality metrics.

Table VI summarizes representative whole-frame results. The light flicker whole condition has high flicker score, and the

TABLE VI
REPRESENTATIVE WHOLE-FRAME QUALITY METRIC RESULTS.

Condition	Flicker Score	Sharpness Score
clean	2.15	458.19
light_flicker_whole	15.64	453.53
light_flicker_background	14.10	454.02
motion_blur_whole	2.16	285.68
mixed_whole	8.98	310.13

TABLE VII
REPRESENTATIVE FIXED FACE ROI METRIC RESULTS.

Condition	Face Flicker	Face Sharpness
clean	2.91	41.89
light_flicker_whole	13.20	44.45
light_flicker_face	12.25	46.23
light_flicker_background	2.91	43.17
motion_blur_whole	2.90	25.90
motion_blur_face	2.90	27.43
motion_blur_background	2.91	41.56

motion blur whole condition has low sharpness score. Clean video remains stable and provides the baseline.

The results also show why multiple metrics are necessary. Light flicker mainly affects brightness variation and temporal instability, but it does not necessarily reduce sharpness. Motion blur reduces sharpness but does not strongly increase flicker. Mixed degradation affects several quality dimensions at the same time. Therefore, a single metric would not be enough to describe all reliability risks.

These results show that whole-frame metrics are effective for identifying global degradation. Fig. 3 visualizes the same trend using the generated whole-frame metric charts.

F. Fixed Face ROI Results

Fixed face ROI metrics provide more detailed analysis of facial degradation. When the face or the whole frame is affected by flicker, the fixed face flicker score increases. However, when only the background is affected by flicker, the face flicker score remains close to the clean condition. This means background flicker does not directly affect the face ROI.

Similarly, when the face or the whole frame is blurred, the fixed face sharpness score decreases. In contrast, background-only blur keeps the face sharpness close to clean. This shows that fixed face ROI metrics can distinguish whether the degradation actually affects the face region.

The results support the main design motivation of SIDA-Guard: whole-frame metrics alone are not enough, and fixed face ROI analysis provides more interpretable information about facial degradation.

This is the key advantage of fixed face ROI analysis. A whole-frame metric may consider background degradation to be important because the background occupies a large area. However, a face-based deepfake detector may be more affected by degradation inside the face. By comparing the same face area under all conditions, SIDA-Guard can separate global degradation from face-specific degradation.

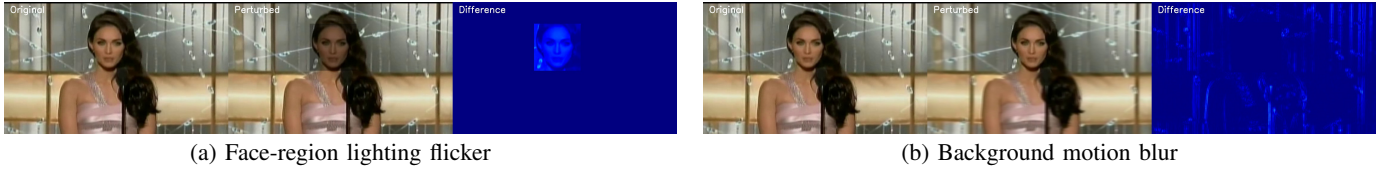


Fig. 2. Visual verification examples using Original, Perturbed, and Difference images. The difference maps verify whether the intended region was modified.

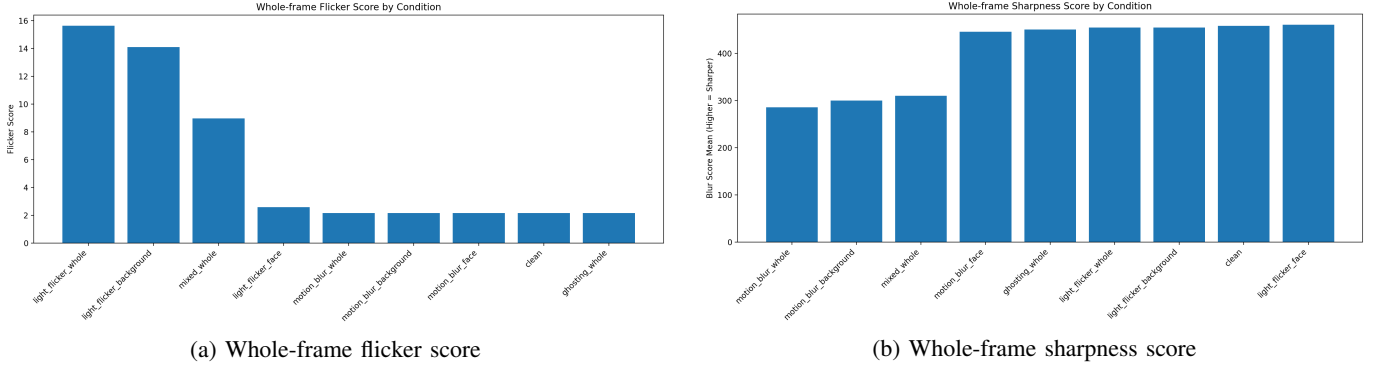


Fig. 3. Whole-frame quality metrics by perturbation condition. Flicker conditions increase brightness instability, while motion blur conditions reduce sharpness.

TABLE VIII
SIDA-GUARD FINAL QUALITY RISK AND RELIABILITY RESULTS.

Condition	Final Risk	Reliability
mixed_whole	0.618	0.382
light_flicker_whole	0.484	0.516
motion_blur_whole	0.300	0.700
light_flicker_face	0.296	0.704
light_flicker_background	0.235	0.765
motion_blur_face	0.172	0.828
motion_blur_background	0.114	0.886
ghosting_whole	0.031	0.969
clean	0.000	1.000

Fig. 4 shows the generated face ROI charts.

G. SIDA-Guard Risk Score Results

The final SIDA-Guard risk score assigns the highest risk to mixed whole-frame degradation. This is reasonable because mixed whole-frame degradation combines flicker, motion blur, and ghosting. In the experiment, mixed_whole obtains a quality risk score of 0.618 and a reliability score of 0.382.

The light_flicker_whole condition receives a risk score of 0.484, showing that strong lighting instability can reduce reliability. Motion_blur_whole obtains a risk score of 0.300. Clean videos receive the lowest risk score of 0.000 and the highest reliability score of 1.000.

These results indicate that SIDA-Guard can convert video quality degradation into an interpretable reliability score. The clean condition is correctly assigned the lowest risk, while the severe mixed degradation condition is assigned the highest risk. Fig. 5 visualizes the final quality risk scores across conditions.

H. Comparison with Published Detector-Centric Work

Published datasets and detectors normally report classification metrics such as accuracy, AUC, or equal error rate. SIDA-Guard is different because it does not claim to improve classification accuracy directly. Instead, it provides an input-quality risk estimate that can be used before or alongside a detector. This means the evaluation is complementary to detector benchmarking.

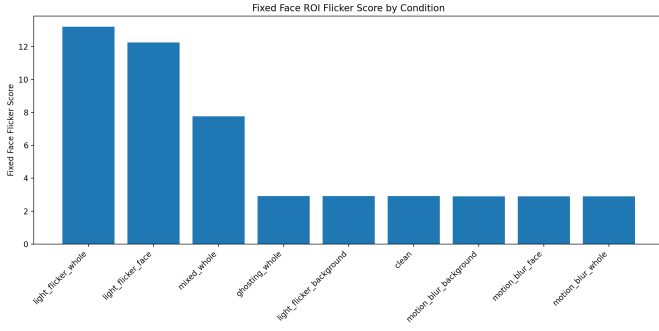
Compared with detector-centric work, the main advantage of SIDA-Guard is interpretability. If a video receives a high risk score, the framework can indicate whether the risk is related to global flicker, global blur, face-region flicker, or face-region blur. This explanation is useful for diagnosing why a detector prediction may be unreliable. In contrast, a detector output alone may provide only a real/fake score without explaining the input quality problem.

The framework is also more lightweight than training a detector. It uses classical computer vision operations and quality metrics instead of large-scale supervised learning. This makes the method easier to reproduce in a limited-time project setting. However, this also means that SIDA-Guard should be treated as a reliability layer, not as a replacement for strong published detectors.

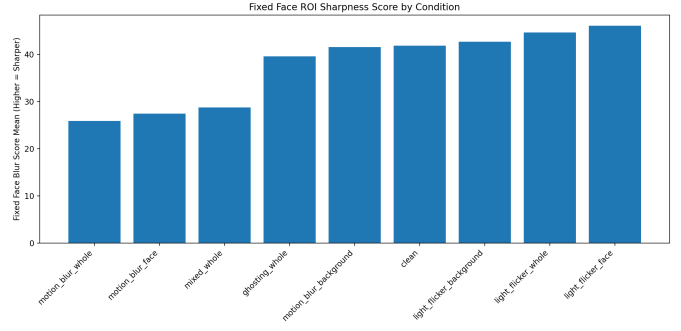
I. Discussion

The experimental results show that SIDA-Guard provides interpretable quality analysis without training a new model. Whole-frame metrics capture global degradation, while fixed face ROI metrics identify whether the facial region is degraded. This is important for deepfake detection because face-based detectors may be more sensitive to facial degradation than background degradation.

However, the current version has several limitations. First, the experiment uses a small sample size because it is a proto-



(a) Fixed face ROI flicker score



(b) Fixed face ROI sharpness score

Fig. 4. Fixed face ROI quality metrics by perturbation condition. Face-region flicker and blur are clearly separated from background-only degradation.

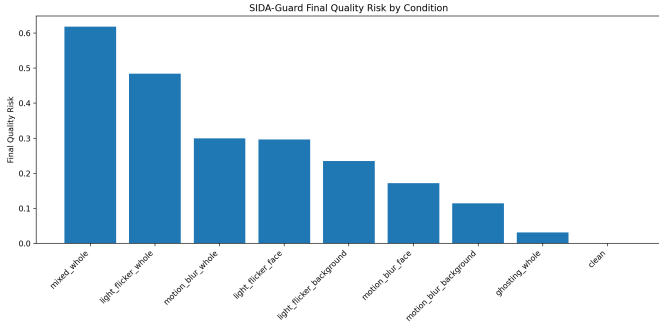


Fig. 5. Final SIDA-Guard quality risk by perturbation condition. Mixed whole-frame degradation receives the highest risk, while clean videos receive the lowest risk.

TABLE IX
COMPARISON BETWEEN SIDA-GUARD AND TYPICAL DETECTOR EVALUATION.

Aspect	Detector-Centric Work	SIDA-Guard
Main output	Real/fake prediction	Quality risk score
Training need	Usually requires training	No detector training
Main concern	Classification performance	Reliability of input quality
Interpretability	Often limited	Metric-based explanation
Future integration	Detector itself	Add-on reliability layer

type. Second, the current implementation does not integrate a formal video deepfake detector. Third, the face detector is simple and may be replaced by stronger models such as RetinaFace or MediaPipe in future work. Fourth, the ghosting perturbation appears weaker than flicker and blur under the current parameter setting, and future experiments can adjust its severity.

Another limitation is that the current risk score is designed as a practical prototype rather than a statistically calibrated uncertainty score. The score is useful for comparing degradation conditions inside this experiment, but future work should validate the score against detector errors on a larger benchmark.

For example, a formal experiment can test whether videos with high SIDA-Guard risk are more likely to cause false positives or false negatives in Xception or other DeepfakeBench models.

Despite these limitations, the results support the feasibility of the proposed idea. The system successfully separates several types of degradation, records traceable metadata, and produces interpretable scores. These properties are valuable because real-world media verification systems need not only strong detectors, but also mechanisms for knowing when detector predictions should be treated with caution.

J. Reproducibility Notes

The project is designed so that another user can reproduce the main pipeline without training a deep neural network. The GitHub repository contains the source code, README file, installation instructions, dataset preparation guidance, and selected generated outputs. The raw video dataset is not stored in the repository because deepfake datasets are large and may have their own access rules. Instead, the README explains how the dataset should be mounted in a Kaggle Notebook environment or provided through a local input directory.

The pipeline searches the input directory recursively for supported video files. It then uses folder and filename keywords to identify real and fake samples. This design is not as strict as a manually curated metadata file, but it makes the prototype easier to use across different dataset structures. If a dataset uses different folder names, the keyword list can be edited in the source code. This is documented in the repository so that future users can adapt the pipeline to FaceForensics++, DFDC, Celeb-DF, or a smaller custom subset.

Reproducibility also depends on preserving intermediate outputs. For this reason, SIDA-Guard saves processed video paths, perturbation metadata, whole-frame metrics, fixed face ROI metrics, preview images, and final risk scores. These intermediate outputs make it possible to inspect whether a high risk score is caused by the intended degradation. For example, if the final risk score is high for mixed whole-frame degradation, a user can inspect the corresponding metadata and preview image to confirm that flicker, blur, and ghosting were actually applied.

K. Threats to Validity

Several threats to validity should be considered when interpreting the results. The first threat is sample size. The current experiment uses a small number of real and fake videos because the project is a time-limited prototype. The purpose is to validate the framework and workflow, not to claim dataset-level generalization. A larger experiment would be required before drawing broad conclusions about all deepfake datasets.

The second threat is perturbation realism. The perturbations are generated using OpenCV-based operations. They are controlled and interpretable, but they are still simulations of real-world degradation. Actual real-world videos may contain more complex combinations of compression artifacts, camera exposure changes, sensor noise, and platform-specific re-encoding. Future work should include naturally degraded videos in addition to synthetic perturbations.

The third threat is detector integration. SIDA-Guard currently estimates quality risk and reliability, but it does not directly evaluate the error rate of a formal deepfake detector. Therefore, the reported risk score should be interpreted as a quality-based warning signal rather than as a calibrated probability of detector failure. The next step is to connect SIDA-Guard to a detector such as Xception or another DeepfakeBench model and measure whether high-risk inputs correlate with false positives or false negatives.

Finally, the current face ROI extraction uses a simple face detection strategy. This is sufficient for demonstrating fixed ROI comparison, but stronger face detectors may improve robustness. If the initial clean face detection is inaccurate, all subsequent fixed ROI metrics may be affected. Future implementations can use stronger detectors, temporal smoothing, and multi-frame face box aggregation to improve the reliability of the fixed face region.

V. CONCLUSION

In this project, we proposed SIDA-Guard, a no-training quality-aware reliability layer for deepfake detection. Instead of training a new detector, SIDA-Guard analyzes whether input video quality may make detector predictions unreliable.

The framework generates controlled perturbations, records metadata, extracts whole-frame and fixed face ROI quality metrics, and computes a final quality risk score. Experimental results show that whole-frame metrics capture global degradation, while fixed face ROI metrics distinguish face degradation from background degradation.

The final risk score provides an interpretable warning mechanism. This allows SIDA-Guard to support existing deepfake detection systems by indicating when predictions should be accepted, treated with caution, or rejected for further verification.

Future work will integrate SIDA-Guard with formal video deepfake detectors such as Xception from DeepfakeBench, evaluate false positive reduction, replace the face detector with a stronger model, and scale the experiment to larger datasets. Another future direction is to calibrate the reliability score using detector confidence and ground-truth error rates.

This would allow SIDA-Guard to become not only a quality analysis tool, but also a detector-aware reliability estimator.

PRACTICAL IMPLICATIONS

SIDA-Guard is intended to support, not replace, existing deepfake detection systems. In a practical workflow, a detector may still be responsible for producing the final real or fake prediction. SIDA-Guard can be used before or alongside that detector to provide a reliability warning. For example, if a video receives a high quality risk score because the face region is strongly blurred or affected by flicker, the system can warn the user that the detector output should be interpreted carefully.

This design is useful in media verification and educational settings because it separates two questions that are often mixed together. The first question is whether the video is fake. The second question is whether the input video quality is good enough for the detector to make a reliable decision. By separating these questions, SIDA-Guard helps avoid overconfidence in predictions made on visually degraded videos.

The proposed framework can also help future researchers debug detector failures. If a detector produces many false positives on real videos, SIDA-Guard metrics can be used to inspect whether those videos contain high flicker, strong blur, unstable frames, or face-region degradation. This makes the method useful as an analysis tool even before it is connected to a complete detector benchmark.

DATA AVAILABILITY STATEMENT

The source code, README file, installation instructions, dataset information, experimental scripts, and selected output results are available in the project GitHub repository:

<https://github.com/CHANGCHENGLIN/sida-guard-deepfake-reliability>

The raw video dataset is not included in the repository because deepfake datasets are usually large and may have their own licenses or access restrictions. This project was tested using sampled real and fake videos from the Celeb-DF v2 dataset in a Kaggle Notebook environment. During the experiment, the dataset was accessed through Kaggle input storage. The pipeline searches `/kaggle/input` recursively for supported video files and detects real/fake samples using folder or filename keywords.

CODE AVAILABILITY

All source code used for video sampling, perturbation generation, metadata recording, quality metric extraction, fixed face ROI analysis, and SIDA-Guard risk scoring is available in the GitHub repository under the `src/` folder:

<https://github.com/CHANGCHENGLIN/sida-guard-deepfake-reliability>

The repository also includes a `README.md` file with installation and execution instructions, a `data/README.md` file with dataset preparation guidance, and selected generated outputs used in the report. The raw video dataset should be

prepared separately by following the dataset provider's access rules.

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REFERENCES

- [1] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Niessner, "FaceForensics++: Learning to detect manipulated facial images," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.
- [2] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The DeepFake Detection Challenge (DFDC) dataset," *arXiv preprint arXiv:2006.07397*, 2020.
- [3] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for deepfake forensics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3207–3216.
- [4] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.
- [5] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: A compact facial video forgery detection network," in *2018 IEEE International Workshop on Information Forensics and Security*, 2018, pp. 1–7.
- [6] L. Li, J. Bao, H. Yang, D. Chen, and F. Wen, "Face X-Ray for more general face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 5001–5010.
- [7] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "DeepfakeBench: A comprehensive benchmark of deepfake detection," *arXiv preprint arXiv:2307.01426*, 2023.